1. The following four waves are sent along strings with the same linear densities (x is in meters and t is in seconds). Rank the tension (τ) in the strings along which they travel, greatest first:

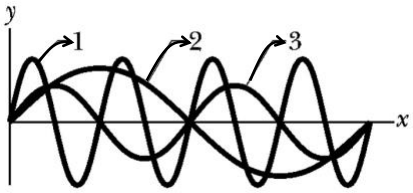
$$y_1 = (3\text{ mm})\sin(x - 3t)$$
$$y_2 = (6\text{ mm})\sin(2x - t)$$
$$y_3 = (1\text{ mm})\sin(4x - t)$$
$$y_4 = (2\text{ mm})\sin(x - 2t)$$

A) $\tau_1 > \tau_4 = \tau_2 > \tau_3$
B) $\tau_3 > \tau_2 = \tau_4 > \tau_1$
C) $\tau_3 > \tau_2 > \tau_4 > \tau_1$
D) $\tau_1 > \tau_4 > \tau_2 > \tau_3$

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2. The figure is a composite of three snapshots, each of a wave traveling along a particular string. The phases for the waves are given by **(a) 2x − 4t, (b) 4x − 8t**, and **(c) 8x − 16t.** Which phase corresponds to which wave in the figure?

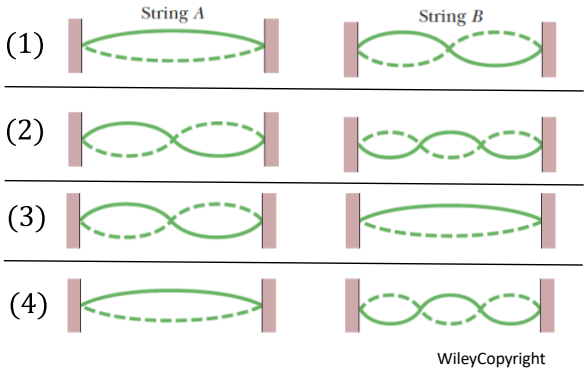
A) (a) to 1; (b) to 2; (c) to 3
B) (a) to 3; (b) to 2; (c) to 1
C) (a) to 2; (b) to 3; (c) to 1
D) (a) to 3; (b) to 1; (c) to 2



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3. Strings A and B have identical lengths and linear densities, but string B is under greater tension than string A. Figure shows four situations, (1) through (4), in which standing wave patterns exist on the two strings. In which situations is there the possibility that strings A and B are oscillating at the same resonant frequency?

A) (1)
B) (2)
C) (3)
D) (4)



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4. Four waves are to be sent along the same string, in the same direction:

$$y_1(x,t) = (4.00\ \text{mm})\sin(2\pi x - 400\pi t)$$
$$y_2(x,t) = (4.00\ \text{mm})\sin(2\pi x - 400\pi t + 0.5\pi)$$
$$y_3(x,t) = (4.00\ \text{mm})\sin(2\pi x - 400\pi t + \pi)$$
$$y_4(x,t) = (4.00\ \text{mm})\sin(2\pi x - 400\pi t + 1.5\pi)$$

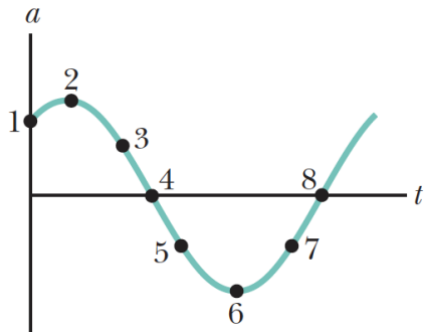
What is the wave function of the resultant wave of these waves?

A) $y'(x,t) = 0$
B) $y'(x,t) = (5.7\ \text{mm})\sin(2\pi x - 400\pi t + 0.5\pi)$
C) $y'(x,t) = (5.7\ \text{mm})\sin(2\pi x - 400\pi t)$
D) $y'(x,t) = (5.7\ \text{mm})\sin(2\pi x - 400\pi t + \pi)$

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5. The acceleration $a(t)$ of a particle undergoing SHM with amplitude x_m is graphed in the figure. At point 5, particle is at:

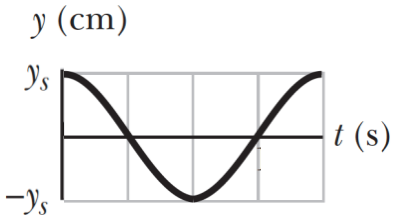
A) $+x_m$
B) $-x_m$
C) Between $-x_m$ and 0
D) Between 0 and $+x_m$



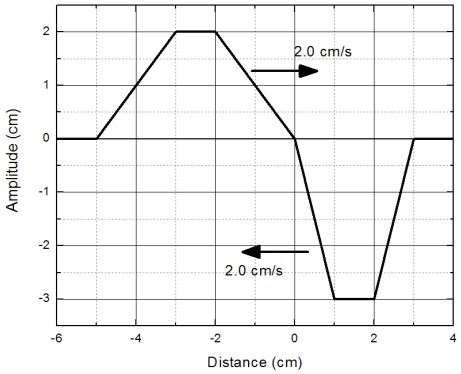
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6. A sinusoidal transverse wave travels along a string in the positive direction of an x axis. The wave equation form is $y(x,t) = y_s \sin(kx - \omega t + \varphi)$. The displacement y of the string particle at $x = 0$ is given in figure as a function of time t. The wave function is to be in the form of:

A) $y(x,t) = +y_s \sin(kx - \omega t)$
B) $y(x,t) = -y_s \sin(kx - \omega t)$
C) $y(x,t) = +y_s \cos(kx - \omega t)$
D) $y(x,t) = -y_s \cos(kx - \omega t)$



7. The graph shows two waves at time t = 0 s, one moving toward the right at 2.0 cm/s and the other moving toward the left at 2.0 cm/s. What will the amplitude be at x = 0 at time t = 1.0 s?

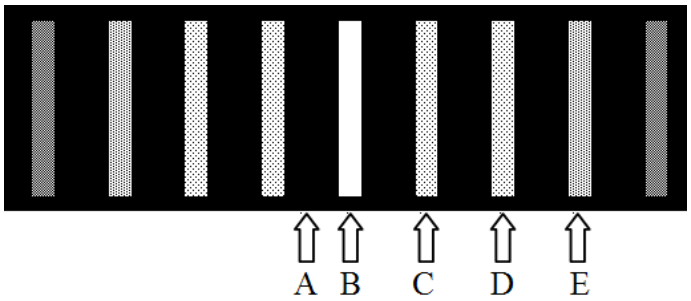


A) −3 cm
B) zero cm
C) –1 cm
D) –2 cm

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8. A beam of monochromatic light with a wavelength of 660 nm is directed at a double slit. Consider the five fringes labeled in the drawing, including the central maximum that is labeled "**B.**" Which one of these fringes is produced when the path difference is 1980 nm?

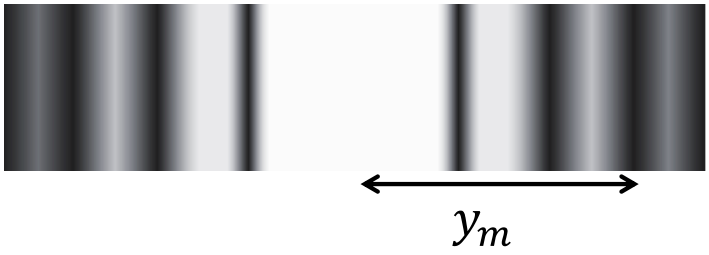
A) A
B) C
C) D
D) E



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9. Light of wavelength 600 nm is incident upon a single slit with width 4×10^{-4} m. The figure shows the pattern observed on a screen positioned 2m from the slit. Determine the distance y_m.

A) 0.003 m
B) 0.006 m.
C) 0.009 m
D) 0.012 m

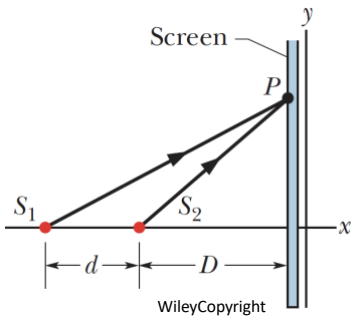


Questions 10 and 11 are related to the following situation:

In the Figure, two isotropic point sources S_1 and S_2 emit light in phase at wavelength λ and at the same amplitude. The sources are separated by distance $d = 6.00\,\lambda$ on an x axis. A viewing screen is at distance $D = 20.0\,\lambda$ from S_2 and parallel to the y axis. The figure shows two rays reaching point P on the screen, at height y_P.

10. What is the path length difference between two rays when $y_p = d$?

A) 3.50 λ
B) 20.0 λ
C) 5.80 λ
D) 26.7 λ



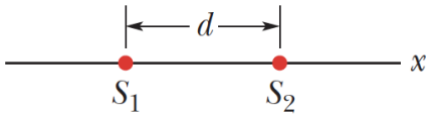
11. If the path length difference between two rays from S_1 and S_2 at P (above figure) equals to 7.4 λ, the resulting intensity at point P is

A) Minimum
B) Maximum
C) Intermediate but closer to minimum
D) Path length difference 7.4 λ for the above situation is impossible

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12. In the Figure, two radiofrequency point sources S_1 and S_2, separated by distance $d = 2.0$ m, are radiating in phase with $\lambda = 0.50$ m. A detector moves in a large circular path around the two sources in a plane containing them. How many maxima does it detect?

A) 10
B) 16
C) 18
D) 20

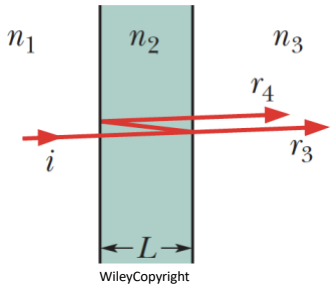


Questions 13 and 14 are related to the following situation:

In the Figure, light is incident perpendicularly on a thin layer of material 2 that lies between (thicker) materials 1 and 3. (The rays are tilted only for clarity.) Part of the light ends up in material 3 as ray r_3(the light does not reflect inside material 2) and r_4(the light reflects twice inside material 2). The waves of r_3 and r_4 interfere, and here we consider the type of interference to be either maximum (max) or minimum (min). For this situation, each following questions below refers to the indexes of refraction n_1, n_2, and n_3, the type of interference, the thin-layer thickness L in nanometers, and the wavelength λ in nanometers of the light as measured in air.

Question 13 find the wavelength that is in the visible range.

Question 14 find the third least thickness. (The visible wavelengths range from 0.4μm to 0.7μm)



Question	n_1	n_2	n_3	Type	L	λ
13.	1.32	1.75	1.39	min	335	

A) 498 nm
B) 480 nm
C) 469 nm
D) 455 nm

Question	n_1	n_2	n_3	Type	L	λ
14.	1.55	1.60	1.33	Min	3rd	512

A) 400 nm
B) 423 nm
C) 478 nm
D) 497 nm

15. A wave in a fluid is incident on a boundary with another fluid. The wave travels from the left, in a fluid with mass density $\alpha\rho$, to the right, into the fluid with mass density ρ (i.e., the mass density of the fluid on the left is α times that of the fluid on the right). The two fluids have the same bulk modulus. Which option gives *power* transmission coefficients (T) at this boundary?

A) $\frac{2\sqrt{\alpha}}{1+\sqrt{\alpha}}$

B) $\frac{4\sqrt{\alpha}}{(1+\sqrt{\alpha})^2}$

C) $\frac{2}{1+\sqrt{\alpha}}$

D) $\frac{4\sqrt{\alpha}}{1+\sqrt{\alpha}}$



16. A whistle of frequency 540 Hz moves in a circle of radius 60 cm at an angular speed of 15.0 rad/s. What is the *lowest* frequency heard by a listener a long distance away, at rest with respect to the center of the circle? (For a circular motion the linear velocity $v = r\omega$, whereas r is the radius of the circle and ω is the angular speed.)

A) 526 Hz
B) 534 Hz
C) 555 Hz
D) 565 Hz

17. Which of the following statements is FALSE about the photoelectric effect?

A) The photoelectric effect is a consequence of the particle character of radiation.
B) There exists a minimum frequency of incident radiation below which photocurrent does not occur.
C) The work function defines the energy needed to detach a photoelectron from the surface of the target material.
D) There exists a lag time where electrons gain sufficient energy to leave the electrode surface.

18. What is a black body?

A) An object painted with black, non-reflective paint
B) An object that does not radiate any electromagnetic waves
C) A model of an object that emits electromagnetic waves with a characteristic frequency distribution
D) All of the above

19. In the Compton effect, at what scattering angle, θ would the shift in the wavelength (i.e., Compton shift) become negligible.

A) $\theta = 90^o$
B) $\theta = 45^o$
C) $\theta = 0^o$
D) $\theta = 15^o$

20. When the Compton shift was measured, what was demonstrated?

A) Photons have momentum.
B) Photons have energy.
C) Photons behave like waves.
D) Photons have no mass.

21. If the frequency of vibration of an atom is f, the atom's energy is

A) Zero
B) 0.5 hf
C) 1.0 hf,
D) 1.5 hf.

22. If the accuracy in measuring the position of a particle increases, the accuracy in measuring its momentum will:

A) increase.
B) decrease.
C) remain the same.
D) It is impossible to say since the two measurements are independent and do not affect each

23. When the photoelectric effect experiments were performed, one effect was inconsistent with classical physics. What was it?

A) The kinetic energy of the ejected electrons did not vary with light intensity.
B) The fact that electrons could form a current within a vacuum.
C) The kinetic energy of the ejected electrons increased as the frequency of light increased.
D) The fact that light could free electrons from the surface of a metal.

24. Photon A has twice the energy of photon B. Which statement is true?

A) the momentum of A is greater than that of B and the wavelength of A is less than that of B
B) the momentum of A is less than that of B and the wavelength of A is greater than that of B
C) the momentum of A is greater than that of B and the wavelength of A is greater than that of B
D) the momentum of A is greater than that of B and the wavelength of A equals that of B

25. Using Stefan's law, determine the total power of blackbody radiation emitted across the entire spectrum of wavelengths for a blackbody with the surface area of 25 m^2 at 60 degrees Celsius.

A) 4.72 W
B) 17.43 kW
C) 18.37 W
D) 7.45 kW

26. What is the magnitude of the momentum associated with a photon having an energy equal to the electron rest energy?

A) $4.73\times 10^{-12}\, m$
B) $2.43\times 10^{-12}\, m$
C) $3.23\times 10^{-12}\, m$
D) $5.12\times 10^{-12}\, m$

The units and the power for the options should be $\ldots \times 10^{-22} kg.\frac{m}{s}$. The correct answer is $2.73\times 10^{-22} kg.\frac{m}{s}$.

27. The position along the *x* axis of an electron is known to be between –0.31 nm and + 0.31 nm. How would the uncertainty in the momentum of the electron change if the electron were allowed to be between –0.62 nm and +0.62 nm?

A) The uncertainty in the momentum would be twice its previous value.
B) The uncertainty in the momentum would be half of its previous value.
C) The uncertainty in the momentum would be four times its previous value.
D) The uncertainty in the momentum would be one fourth its previous value.

28. What percentage increase in wavelength leads to a 75% loss of photon energy in a photon-free electron collision?

A) 200 %
B) 300 %
C) 400 %
D) 500 %

29. If you double the kinetic energy of a nonrelativistic particle, how does its de Broglie wavelength change?

A) $1/2$
B) 2
C) $\sqrt{2}$
D) $1/\sqrt{2}$

30. The wave functions for an electron in an infinite, one dimensional potential well with length L along an x axis is given by $\psi(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{\pi}{L}x\right)$. What is the probability of finding the electron between $x = (0.50)L$ and $x + dx = (0.51)L$?
A) 0.014
B) 0.005
C) 0.020
D) 0.050